RAFFLES INSTITUTION
H2 Mathematics (9758)
2026 Year 6

Additional Practice Questions for Chapter S4: Sampling

1 With the rising number of students owning a mobile device, Nurturing Primary School decides to conduct a survey on the amount of screen time the students spend on their mobile devices each day. There are 8 classes for each level from Primary One to Primary Six.

(a) As Mrs Neha only teaches the Primary Six cohort, she decides to conduct the survey for all the 8 classes of Primary Six students. Explain whether these 8 classes of Primary Six students form a sample or a population. [1]

(b) For the data collected to be more representative of the entire student population, Mrs Neha decides to conduct the survey for 100 students from Primary One to Primary Six. She randomly surveyed 100 students who stayed back in school for CCA activities on a particular day. Explain whether this sample is a random sample. [1]

2 A manufacturer of candles claimed that it produced birthday candles with a mean burning time of 6 minutes. A random sample of 150 birthday candles was tested and the burning times, X minutes, were summarized by $\sum(x-5)=120$ and $\sum(x-5)^2=638$. Calculate the unbiased estimates for the mean μ and variance σ^2. [3]

[5.8, 3.64]

3 Two firms A and B manufacture similar components with mean breaking strengths of 6 units and 5.5 units, and standard deviations 0.4 units and 0.25 units respectively. It is given that both distributions are normal. If random samples of 100 components from firm A and 50 from firm B are tested, find the probability that the mean breaking strength of the sample from firm A minus that from firm B will be between 0.45 and 0.55 units. [3]

[0.651]

4 The life, in hours, of a randomly chosen light bulb produced by a manufacturer is normally distributed with mean 1100 and standard deviation 70. How large a sample is required such that the probability that the mean life in the sample shall exceed 1120 is not more than 5%? [5]

Do you need to use Central Limit Theorem in your working? Explain. [1]

[least n = 34]

5 The times taken by two runners A and B in a 400 metre race are independent and may be assumed to be normally distributed. The times (in seconds) taken by A has mean 46 and standard deviation 0.5 and the times taken by B has mean 46.2 and standard deviation 0.8.

Let $\overline{X}$ denote the average time for A to run the 400 metre track on 10 different occasions and $\overline{Y}$ denotes the average time for B to run the same track on 16 different occasions. Find the value of t such that

$$\mathrm{P}\left(\left|\overline{X}-\overline{Y}+0.2\right|>t\right)=0.1\,. \qquad [4]$$

[0.419]

6 A sample of n observations was taken from a population of mean μ and variance 10. Find the least sample size required such that the probability of the sample mean lying between $\mu-0.5$ and $\mu+0.5$ is more than 0.89. State an assumption that you have to make in order to proceed with this calculation. [6]

[103]

7 In a certain examination with a very large entry, the marks obtained by the male candidates were found to follow a normal distribution with a mean of 54 and a standard deviation of 16. $\overline{X}$ denotes the mean mark scored by a sample of 4 male candidates.

(i) State the sampling distribution of $\overline{X}$.

(ii) Find the probability that $\overline{X}$ will exceed 70.

(ii) Given that there is a probability of 0.95 that $\overline{X}$ differs from its mean mark by less than c, find the value of c.

In the same examination the marks obtained by the female candidates were found to follow an independent normal distribution with a mean of 59 and a standard deviation of 20.

(iv) Find the probability that the total marks obtained by a randomly chosen male candidate and a randomly chosen female candidate exceed 120.

It is given that $\overline{Y}$ denotes the mean mark scored by a random sample of 5 female candidates.

(v) State the sampling distribution of $\overline{Y}-\overline{X}$.

(vi) Find $\mathrm{P}(\overline{Y}>\overline{X})$.

[**(i)** $\mathrm{N}\left(54,\ 8^2\right)$ **(ii)** 0.0228 **(iii)** 15.7 **(iv)** 0.392 **(v)** $\mathrm{N}\left(5,\ 144\right)$ **(vi)** 0.662]

8 Mike plays a game at a fun-fair. Each game costs \$3 to play. The probability distribution of the amount of money \$$X$, that Mike wins from a game is given by

$$\mathrm{P}(X=x)=\begin{cases} kx & \text{for } x=1,\ 2,\ 3, \\ \frac{1}{3}k(x-1) & \text{for } x=4,\ 5,\ 6, \\ 0 & \text{otherwise,} \end{cases}$$

where k is a constant.

(i) Show that $k=\dfrac{1}{10}$. [1]

(ii) Find the expectation and variance of his net gain for one game. [3]

(iii) Assuming that each game is independent of one another, estimate the probability that his total net gain in 30 games is more than \$20. [3]

[**(ii)** $\dfrac{7}{15}, \dfrac{566}{225}$ **(iii)** 0.245]

9 During each round of practice, John, does 10 multiple-choice questions and his score X, is the number of questions he answered correctly. On average, he has an 85% chance of answering each multiple-choice question correctly.

(i) State in the context of the question, one assumption needed to model X by a binomial distribution. [1]

On a particular day, John does 4 rounds of practice.

(ii) Find the probability that the total score for John in the 4 rounds of practice is more than 36. [2]

(iii) Find the probability that John obtains a score of at most 9 in each of his first 2 rounds of practice. [3]

To qualify for an award, John will need a mean score of at least 8.8 for all his 50 rounds of practice.

(iv) Estimate, using an approximation, the probability that John will qualify for the award. [3]

[**(ii)** 0.130 **(iii)** 0.645 **(iv)** 0.0301]

10 (a) Anne, a Bubble Tea (BBT) seller, intends to increase the sales of BBT using a drink vending machine which delivers BBT into a cup when cash payment is made into the machine.
The volume of BBT dispensed is normally distributed with mean 210 ml and standard deviation 5 ml. The capacity of a cup is 220 ml and the nominal amount of BBT in a cup is stated as 212 ml.

(i) Find the probability that a cup overflows when BBT is dispensed into the cup. [1]

(ii) A customer bought five cups of BBT from the vending machine. Find the probability that at most one cup of BBT will overflow. [2]

(iii) Anne received complaints from some customers that there is a high proportion of cups with less than the nominal amount of BBT. It is assumed that the standard deviation of the volume of BBT dispensed is fixed while the mean volume of BBT dispensed could be adjusted. Find the range of the mean volume of BBT dispensed such that not more than 10% of the cups will contain less than the nominal amount of BBT. [3]

(iv) Another customer bought n cups of BBT. Find the approximate probability that the total volume of BBT dispensed exceeds $211n$ ml as n becomes very large. [2]

(v) Anne wishes to gather feedback about her BBT. She decided to interview 50 customers who bought BBT from the vending machine during lunch time. Give a reason as to whether Anne would obtain a random sample of customers. [1]

(b) On another occasion, Anne deployed a staff to operate a BBT counter at a wedding reception. The average volume of BBT per serving is 200 ml and the standard deviation of the volume of BBT is given to be 10 ml. Estimate the probability that the mean volume of 60 servings of BBT prepared by the staff is less than 198 ml. [3]

[**(a)(i)** 0.0288 **(ii)** 0.995 **(iii)** [218, 220] **(iv)** 0 **(b)** 0.0607]

THE END